CENG381 Stochastic Processes

Chapman-Kolmogorov & Diagonalization — Practice 1

Q1 (30 points). A 2-state weather Markov chain has states {S = Sunny, R = Rainy} with daily transition matrix:

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          S      R
S  [   0.7    0.3  ]
R  [   0.5    0.5  ]
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a) Compute $P^2 = P \cdot P$ directly by matrix multiplication. Show all arithmetic.

b) State the Chapman-Kolmogorov equation for P^(n+m)(i,j) in terms of P^n and P^m. Using P^2 from part (a), apply C-K to compute $P^4 = P^2 \cdot P^2$.

c) Verify one entry: show that P^4(S,S) from part (b) equals the (S,S) entry you get from multiplying $P^2 \cdot P^2$ directly.

Q2 (35 points). Continuing with the same chain (p = 0.3, q = 0.5, so p+q = 0.8). Use diagonalization to derive a closed-form expression for P^n.

a) Find the eigenvalues of P by solving $\det(P - \lambda I) = 0$. Show that $\lambda_1 = 1$ and $\lambda_2 = 1 - p - q = 0.2$.

b) For a 2-state chain with P(S,S)=1−p and P(R,S)=q, the n-step return probability is:

P^n(S,S) = q/(p+q) + (1−p−q)^n · p/(p+q)

Verify this formula for n=1 by checking it recovers P(S,S)=0.7.

c) Use the formula to compute P^4(S,S). Confirm it matches the result from Q1(b).

Q3 (35 points). Still using the weather chain (p=0.3, q=0.5). The closed-form P^n(S,S) = 5/8 + (3/8)·(0.2)^n allows us to study long-run behaviour precisely.

a) Find the stationary distribution π by taking the limit P^n(S,S) → π_S as n → ∞. Verify π by solving $\pi \cdot P = \pi$.

b) Find the smallest integer n* such that |P^n(S,S) − π_S| < 0.01. Show your work using the formula.

c) The second eigenvalue $\lambda_2 = 0.2$ governs the speed of convergence. Suppose instead $\lambda_2 = 0.9$ (a nearly-periodic chain). How would n* change? Explain qualitatively why chains with λ_2

closer to 1 mix more slowly.